

Direct On Line [DOL] Motor Very Basic MAPS Template (Electrical)

The Very Basic Direct On Line (DOL) Motor MAPS template is designed to control a motor that is connected as a DOL.

All MAPS templates provide the following:

- an associated PLC function block
- a set of SCADA tags (Adroit agents) linked to the provided signals
- an associated SCADA graphic with inbuilt faceplates

This document describes the Very Basic DOL Motor MAPS Template, which provides the lowest level of control over the DOL motor. For details of its required PLC memory usage; number of PLC I/O and SCADA size (scan points) - see the **Resources** table below.

Very Basic DOL Motor Functional Philosophy

The DOL start output is enabled once all interlocks are ready and a start request is received.

Field Interlocks

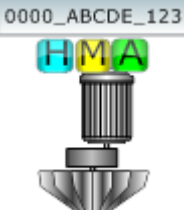
- Trip Feedback
- Fault Feedback

Interlock fault recovery process – once a field interlock is activated the faceplate will display a red indication, if the interlock is recovered and still latched in the PLC function block and the faceplate will show a green indication and the reset button on the faceplate will flash, pressing the Reset button will unlatch the interlock and display a grey indication and allow the DOL to become ready for operation.

Start and stop requests are initiated by selecting a mode on the DOL faceplate:

- **Auto mode** - the auto control program initiates a start and stop, normal PLC control.
- **Manual mode** - the operator can start and stop from the faceplate.
- **Hand mode** - the motor is in PLC bypass mode.

Typical Faceplate Graphics for the Very Basic DOL Motor MAPS Template

DOL ISO	DOL Aerator	DOL Standard Pump Side View Left	DOL PUMP LEFT	DOL PUMP RIGHT
				
	Hand Mode			
	Manual Mode			
	Auto Mode			
	Motor Control Name			
Graphics legend:				
Line colour	Fill colour	DOL status		
Black	Grey	Stopped		
Grey	White	Stopping		
Green	White	Starting		
Black	Green	Running		
Black	Yellow/Red	Fault Active		

Signal Description		Agent Type	Very Basic
SCADA Control Words:			
SSCL 0	Control Word	Marshal	x

Signal Description		Agent Type	Very Basic
SCADA Status Words:			
SSSL 0	Status Word	Marshal	x
SSSL 1	Total Running Hours	Analog	x

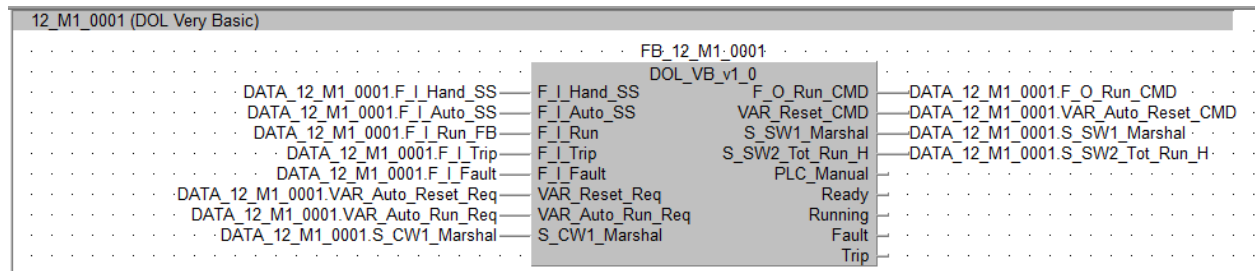
Digital Inputs:		Very Basic
DI 0	Hand (On = Hand)	x
DI 1	Auto (On = Auto)	x
DI 2	Run Feedback (On = Running)	x
DI 3	Trip (On = Tripped)	x
DI 4	Fault (On = Fault)	x
Digital Outputs:		
DO 0	Device Start	x

SCADA Control Word (Adroit Marshal Agent)		Event Logged	Very Basic
Bit 0	Alarms All Reset	x	x
Bit 1	Manual	x	x
Bit 2	Start Command	x	x
Bit 3	Stop Command	x	x
Bit 4	Spare		
Bit 5	Spare		
Bit 6	Spare		
Bit 7	Spare		
Bit 8	Spare		
Bit 9	Spare		
Bit 10	Spare		
Bit 11	Spare		
Bit 12	Spare		
Bit 13	Spare		
Bit 14	Spare		
Bit 15	Spare		

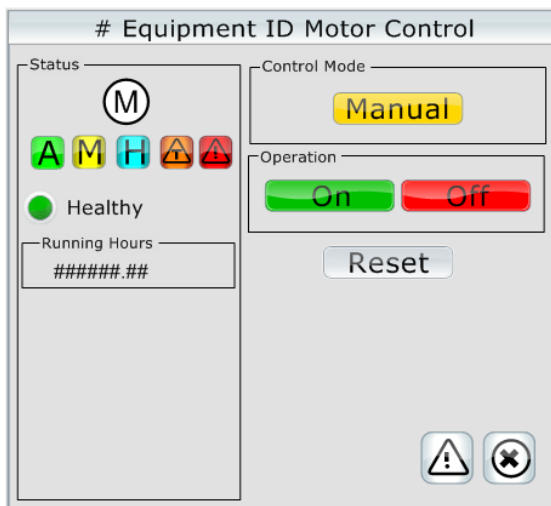
SCADA Status Word (Adroit Marshal Agent)		Alarmed	Very Basic
Bit 0	Hand		x
Bit 1	Auto		x
Bit 2	Manual		x
Bit 3	Ready		x
Bit 4	Starting		x
Bit 5	Running		x
Bit 6	Trip	x	x
Bit 7	Fault	x	x
Bit 8	Spare		
Bit 9	Spare		
Bit 10	Spare		
Bit 11	Spare		
Bit 12	Spare		
Bit 13	Spare		
Bit 14	Spare		
Bit 15	Spare		

Resources	Very Basic
Digital Inputs	5
Digital Outputs	1
Analog Inputs	0
Analog Outputs	0
PLC Memory Steps	178
System Words Used	9
System Bits Used	66
SCADA SCAN Points	2

Template: Very Basic DOL PLC Function Block



Template Graphic: Home Screen



Control



Descriptions

Manual control by the operator

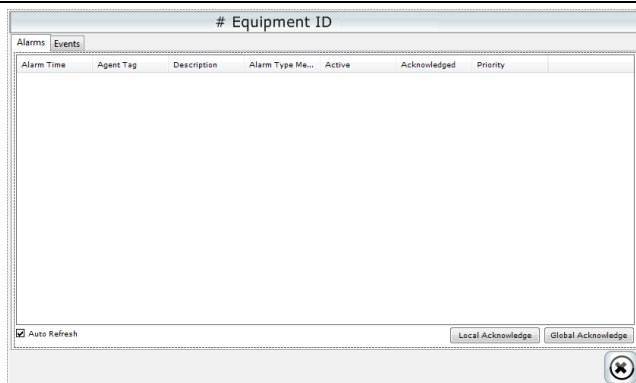
Manually start the DOL

Manually stop the DOL

Reset the DOL after a fault has been repaired

Healthy to start the DOL

Template Graphic: Alarms & Events Screen



Control

Alarms

Display all the alarms filtered for the current Digital Output

Events

Display all the events filtered for the current Digital Output.

Local Acknowledge

Acknowledge all the alarms on locally on the current machine

Global Acknowledge

Acknowledge all the alarms on globally on all the machines.